Growing Food in the Dragoon Mountains

A Complete Hydroponic Guide

Water Treatment · Locally-Sourced Nutrients · System Design
Seasonal Calendar · Crop Selection · pH Management

A Practical Guide for High-Desert Hydroponic Growing · Cochise County, Arizona

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1 Know Your Environment

The Dragoon Mountains in Cochise County, southeastern Arizona, are a sky island range rising to 7,500 ft. This unique geography gives you significant advantages over low-desert growing — cooler summers, reliable monsoon rains, and rich local mineral resources — but also demands specific strategies.

Climate Overview

Factor	Conditions	Impact on Hydroponics
Elevation	4,500 – 7,500 ft	Stronger UV, lower air pressure, cooler than low desert
Summer highs	85 – 95°F	Manageable vs. Phoenix 110°F+ — plants stay healthier
Night temps	30 – 40°F swing	Large swings stress plants — use insulated reservoir
Monsoon season	July – September	50–70% humidity; ideal soft rainwater collection opportunity
Annual rainfall	15 – 20 inches	Mostly monsoon — plan catchment infrastructure around it
Winter frost	Nov – Feb at elevation	Protected structure needed; great for cold-tolerant greens

Your Key Advantages

- Cooler summers than low-desert Arizona — fruit crops don't abort flowers from heat stress
- Monsoon season delivers soft, low-TDS rainwater — ideal as reservoir source water
- Rich mineral geology: iron-bearing granite, caliche calcium, sulfur deposits, copper ores
- Abundant oak woodland provides potassium (ash) and chelating tannins (bark/leaves)
- Bat caves throughout Cochise County supply high-nitrogen, high-phosphorus guano
- Long, warm fall season is your best growing window — take full advantage

> **Tip:** Fall (September–November) is your prime growing season in the Dragoons. Plan your biggest plantings to mature during this window.

2 Water Treatment

Water quality is the single most important factor in hydroponic success in southeastern Arizona. SE Arizona well water is typically hard — high in dissolved calcium, magnesium, and especially bicarbonates, which constantly push pH upward. Understanding and treating your source water before adding nutrients is essential.

Step 1 — Test Your Source Water

Before mixing any nutrients, you must know what you're starting with. Two meters are non-negotiable purchases for this system:

- **EC/TDS meter** — measures dissolved minerals already in your water. SE AZ well water commonly reads 200–500 ppm before any nutrients are added.
- **pH meter** — measures acidity/alkalinity. Target 5.8–6.2 for most crops. Local well water often reads 7.5–8.5.

> **Important:** If your source water EC is already 0.3 or higher, reduce your nutrient dose proportionally. Plants don't know the difference between minerals in the water and minerals you added — excess causes toxicity.

Step 2 — The Bicarbonate Problem

Bicarbonates (HCO_3^-) are the main culprit in hard water pH instability. Even after you adjust pH down, bicarbonates act as a buffer and push it back up within hours. The fix is to neutralize the bicarbonates first, before adding nutrients:

Step	Action	What Happens
1	Fill reservoir with source water	Starting pH likely 7.5–8.5
2	Add acid slowly while stirring (see Chapter 5)	Bicarbonates react — water fizzes and releases CO_2
3	Wait for fizzing to stop completely	Bicarbonates are neutralized
4	Adjust pH to 5.8–6.0	Now pH is stable — won't bounce back quickly
5	Add nutrients in correct order	Nutrients dissolve into pre-treated, stable water

Best Water Sources — Ranked

Source	TDS	pH	Treatment Needed	Rating
Monsoon rainwater (collected)	< 20 ppm	6.0–6.5	Minimal — just pH fine-tune	✓✓✓ Best
Filtered/RO well water	< 50 ppm	6.5–7.0	Light pH adjustment	✓✓ Good
Untreated well water	200–500 ppm	7.5–8.5	Full bicarbonate scrub + pH down	■ Usable

Surface/stock tank water	Variable	Variable	Test first — may contain contaminants	■ Test first

Tip: Set up a monsoon water catchment system — even a 500 sq ft roof collects 3,000+ gallons per monsoon season. Store in dark, covered tanks to prevent algae.

3 System Design & Setup

The right system design for the Dragoon Mountains addresses three local challenges: intense UV radiation, large temperature swings, and high evaporation rates. Simplicity and passive resilience are more important here than high-tech solutions.

Recommended Systems

System	Best For	Dragoon Advantage	Key Requirement
Kratky (passive)	Greens, herbs, lettuce	No electricity, no pump — ideal off-grid	Check water level weekly
DWC (Deep Water Culture)	Tomatoes, peppers, cucumbers	Simple, high yield, easy temp management	Air pump + airstone
Drip / Dutch bucket	Large fruiting crops	Best water efficiency for precious water	Timer + pump
Wicking	Herbs, small plants	Zero electricity, completely passive	Correct media ratio

Reservoir Management

- **Bury or insulate reservoirs** — underground tanks stay 10–15°F cooler, keeping root zone in the ideal 65–72°F range
- **Use dark, opaque containers** — prevents algae growth and reduces heat absorption
- **Cover all reservoirs completely** — evaporation in the high desert is extreme; an uncovered 50-gallon reservoir can lose 5+ gallons per day in summer
- **UV-stabilized plastic only** — standard plastic degrades rapidly at Dragoon elevations

Shade & Protection

- **30–40% shade cloth** for summer growing — high-altitude UV bleaches leaves and heats reservoirs
- **East-facing orientation** preferred — morning sun, afternoon shade from the mountains
- **Local ramada structure** using native juniper or oak poles with shade cloth roof — low cost, blends with landscape
- **Greenhouse for winter** — even a simple hoophouse extends your season through frost months

Growing Media Options

Media	Best Use	Local Source?
Perlite	Most systems — excellent drainage, reusable	Purchase — no local equivalent
Coarse granite sand	Wicking, drip systems	✓ Crush local granite — free and abundant

Gravel (washed)	Dutch bucket, large systems	✓ Local creek gravel — wash thoroughly
Coconut coir	Water retention in hot weather	Purchase
Pumice	Excellent drainage + aeration	✓ Volcanic pumice found in SE Arizona geology

> **Tip:** Crushed local granite and washed creek gravel are excellent free growing media. Rinse thoroughly, soak in dilute vinegar to remove carbonates, rinse again, then they are ready to use.

4 Locally-Sourced Nutrients

The Dragoon Mountains and surrounding Cochise County contain nearly every element needed to build a complete hydroponic nutrient solution. The target is to replicate the Masterblend 4-18-38 three-part system: a base nutrient, a calcium source, and magnesium sulfate — plus a full micronutrient profile.

Target Nutrient Profile (per gallon)

Macronutrient	Target ppm	Micronutrient	Target ppm
Nitrogen (N)	116	Iron (Fe) chelated	2.4
Phosphorus (P)	47	Boron (B)	1.2
Potassium (K)	188	Manganese (Mn)	1.2
Calcium (Ca)	113	Zinc (Zn)	0.3
Magnesium (Mg)	41	Copper (Cu)	0.3
Sulfur (S)	58	Molybdenum (Mo)	0.06

Part A — Base Nutrient (K, P, N + Micronutrients)

Nutrient	Local Source	Production Method
Potassium (K)	Oak wood ash	Burn seasoned oak. Steep ash in rainwater 24 hrs. Strain through cloth. Use liquid.
Phosphorus (P)	Bat guano (Cochise County caves)	Collect guano. Steep 1 cup per gallon of water. Aerate vigorously 24–48 hrs. Strain finely.
Nitrogen (N)	Aerated guano tea	Continued aeration converts ammonium to nitrate via nitrifying bacteria. Use within 48 hrs.
Iron (Fe) chelated	Red granite clay + oak bark/leaves	Brew strong oak bark tea. Soak iron-rich red clay in it 48 hrs. Strain. Tannins chelate the iron.
Copper (Cu)	Bisbee area malachite / azurite	Dissolve small chip in dilute apple cider vinegar. Dilute 1:100 before reservoir use.
Boron (B)	Desert caliche, Sulphur Springs soil	Boron concentrates naturally in AZ arid soils. Brew strong caliche soil tea. Use sparingly.

Part B — Calcium Source

Caliche (calcium carbonate hardpan) is abundant throughout the Sulphur Springs Valley east of the Dragoons. To make plant-available calcium:

- Collect caliche chunks from exposed hardpan layers
- Dissolve in apple cider vinegar (from fermentation) → produces calcium acetate, which is fully plant-available
- Combine with nitrified guano tea for calcium + nitrogen together

• Alternative: dolomite limestone dissolved in dilute acid provides both Ca and Mg simultaneously

Part C — Magnesium + Sulfur

The Sulphur Springs Valley is named for its mineral deposits. Look for epsomite — white crystalline crusts around dry springs and mineral seeps. Epsomite is magnesium sulfate in its natural form, identical to commercial Epsom salt.

Micronutrient Production

Nutrient	Local Source	Method
Manganese (Mn)	Dark metamorphic rock, streambed sediment	Collect mineral-stained streambed mud. Brew as dilute tea. Strain before use.
Zinc (Zn)	Tombstone / Bisbee zinc-bearing ores	Dissolve zinc ore chip in dilute acid. Heavily dilute before adding to reservoir.
Molybdenum (Mo)	Wild legume root nodules (mesquite, clover, palo verde)	Ferment nodules from local wild legumes in water. Provides trace Mo amounts.

Important: Copper and boron are toxic to plants at high concentrations. Always make a very dilute stock solution and add drop by drop, testing EC between additions.

5 pH Management

Maintaining pH in the range of 5.8–6.2 is critical for nutrient availability. Outside this range, even perfectly formulated nutrients become locked out and unavailable to plant roots. Both pH up and pH down can be produced locally.

pH Down (Acid) — Lowering pH

Source	How to Make	Strength	Notes
Prickly pear vinegar (best local option)	Ferment crushed prickly pear juice open to air 2–4 weeks	pH 3–4	Unlimited local supply. Gentle, easy to dose.
Fermented citric acid	Ferment honey/prickly pear juice with Aspergillus niger mold 7–10 days	pH 2–3	More concentrated. How industrial citric acid is made.
Sulfur burn method	Burn local sulfur, bubble gas through water outdoors	pH 1–2	Strong — do outdoors only. Fumes are irritating.
Food-grade citric acid (purchase)	Dissolve powder in water to make stock solution	Very precise	Best option if purchasing one item.

pH Up (Alkaline) — Raising pH

Source	How to Make	Strength	Notes
Oak ash lye (best local option)	Pack ash in container, pour rainwater through, collect dark liquid	pH 11–13	Also adds potassium. Dilute heavily before use.
Hydrated lime (from caliche/limestone)	Burn limestone in hot fire, add small amount of water to quicklime	pH 12+	Very strong. Tiny amounts only.
Baking soda (purchase)	Dissolve in water — ready to use	Gentle	Easy to dose. Sodium accumulates over time.

pH Dosing Protocol

- Always add pH adjusters **drop by drop** — small amounts move pH dramatically
- Stir thoroughly and **wait 10 minutes** before re-testing — pH drifts after mixing
- Make a **dilute stock solution** rather than adding raw materials directly to the reservoir
- For well water: pre-treat and neutralize bicarbonates **before** adding any nutrients
- Check pH **daily** for the first week of a new reservoir to understand your water's drift rate

> **Tip:** Keep a small spray bottle of dilute prickly pear vinegar (pH ~4) next to your reservoir for quick daily micro-adjustments. Far easier than dosing from a large bottle.

6 Seasonal Growing Calendar

The Dragoon Mountains have four distinct growing windows. Unlike low-desert Arizona, your elevation gives you a true cool season and a manageable summer. Plan your crops around these windows to maximize yield with minimum intervention.

Season	Months	Conditions	Best Crops	Key Tasks
Spring	Mar – May	Warming, low humidity, mild nights	Tomatoes, peppers, cucumbers, basil	Start seedlings indoors. Begin monsoon water collection prep.
Monsoon	Jul – Sep	Hot days, humid nights, afternoon storms	Basil, herbs, beans, heat-tolerant greens	Collect rainwater. Watch for fungal issues. Reduce nutrients slightly.
Fall ★ Best	Sep – Nov	Perfect temps, clear skies, low humidity	Everything — tomatoes, peppers, lettuce, kale, chard	Your prime season. Plant heavily. Use collected monsoon water.
Winter	Nov – Feb	Cool to cold, frost risk at elevation	Spinach, arugula, kale, mache, cilantro	Protect with hoophouse. Excellent for cold-tolerant greens.

Monthly Action Guide

Month	Key Actions
January	Grow cold-tolerant greens under cover. Plan spring seed orders.
February	Start tomato and pepper seeds indoors. Clean and prep reservoirs.
March	Transplant tomatoes/peppers. Set up shade cloth for summer.
April	Peak spring growing. Monitor pH and EC daily as temps rise.
May	Install shade structures. Insulate or bury reservoirs before heat.
June	Reduce plantings. Focus on heat-tolerant herbs. Pre-monsoon prep.
July	Monsoon begins — start water collection. Watch for fungal issues.
August	Continue water collection. Harvest summer crops. Start fall seeds.
September	Fall season begins — best growing period. Plant everything.
October	Peak fall harvest. Plant winter greens. Fill water storage tanks.
November	Frost protection in place. Transition to cold-season crops.
December	Winter greens under cover. Review season, order seeds for spring.

7 Crop Selection

Crop selection for the Dragoon Mountains should prioritize heat tolerance in summer, cold tolerance in winter, and drought-resilient varieties developed for the Southwest. Your elevation gives you access to a wider variety than low-desert growers.

Top Crops by Category

Crop	Variety	Season	Notes
Cherry Tomato	Solar Fire, Sungold, Heatmaster	Spring / Fall	Heat-tolerant varieties set fruit even at 95°F
Chile Pepper	Hatch, Ancho, Chiltepin (native)	Spring – Fall	Chiltepin is a native Cochise County wild chile — thrives here
Bell Pepper	Karma, Corno di Toro	Fall	Best in fall — avoids summer heat stress
Cucumber	Diva, Marketmore	Spring / Monsoon	Loves humidity — excellent during monsoon season
Kale	Lacinato, Red Russian	Fall / Winter	Becomes sweeter after frost — excellent cold-season crop
Spinach	Bloomsdale, Space	Winter	Thrives in cool weather — ideal hoophouse crop
Arugula	Wild or cultivated	Fall / Winter / Spring	Fast growing, 30-day harvest, bolt-resistant in cool weather
Lettuce	Oak Leaf, Butterhead, Romaine	Fall / Winter / Spring	Avoid summer — bolts quickly above 80°F
Basil	Genovese, Thai, Holy	Summer / Monsoon	Loves the heat and monsoon humidity
Cilantro	Slow-bolt varieties	Fall / Winter / Spring	Bolts in summer — plant in cool seasons only
Oregano	Greek, Mexican	Year-round	Drought-tolerant, thrives at altitude — excellent here
Epazote	Native variety	Summer / Fall	Traditional SW herb, native to region, heat-tolerant

Tip: Chiltepin chile is native to Cochise County and the surrounding sky island ranges. It is extraordinarily well adapted to your exact conditions — consider it your signature local crop.

8 Troubleshooting

Most hydroponic problems in this region fall into three categories: pH drift (caused by hard water bicarbonates), heat-related root stress, and nutrient deficiencies from imprecisely sourced local inputs.

Problem Diagnosis Guide

Symptom	Likely Cause	Local Fix
Yellow leaves between green veins	Iron deficiency (most common)	Brew fresh oak tannin + red clay chelated iron. Apply as foliar spray first for fast response.
pH rising constantly	Bicarbonates in well water	Pre-treat water with prickly pear vinegar before adding nutrients. Full bicarbonate scrub.
Stunted growth, dark green leaves	Phosphorus deficiency	Brew stronger aerated bat guano tea. Ensure aeration is vigorous enough for nitrification.
Brown leaf edges, tip burn	Calcium deficiency or salt burn	Add more caliche-vinegar extract. Check EC — may be too high from hard water. Flush and restart.
Wilting despite wet roots	Root rot from warm reservoir	Insulate or bury reservoir. Add air stone. Target root zone below 72°F.
White crusty deposits on media	Salt/mineral buildup from hard water	Flush with clean rainwater. Switch to monsoon water collection as primary source.
Powdery white coating on leaves	Powdery mildew — monsoon humidity	Increase airflow. Spray dilute apple cider vinegar on leaves. Remove affected leaves.
Slow growth in summer	Heat stress above 95°F	Add 30–40% shade cloth. Bury reservoir. Consider pausing fruiting crops — grow heat-tolerant herbs instead.
Nutrient solution turns cloudy	Bacterial growth from organic inputs	Replace reservoir. Filter guano/organic teas more finely. Use within 48 hrs of brewing.

Essential Tools Checklist

Tool	Why It Matters
EC/TDS meter	Non-negotiable — know your water and nutrient concentration
pH meter	Non-negotiable — the most important daily measurement
Thermometer	Monitor reservoir and air temps — root zone must stay below 72°F
Fine mesh strainer / cloth	Filter all locally-brewed teas before adding to reservoir
Dark covered containers	For reservoirs — prevents algae and UV degradation
Shade cloth (30–40%)	Essential for summer UV protection
Rainwater catchment	Cistern + gutters for monsoon collection — your best water source

pH calibration solution	Keep meter accurate — recalibrate monthly

The Dragoon Mountains offer an exceptional environment for hydroponic growing. Your monsoon rains, local mineral wealth, cooler elevation temperatures, and long fall season are genuine advantages. With the right water treatment and locally-sourced nutrients, a highly productive system is achievable with minimal purchased inputs.